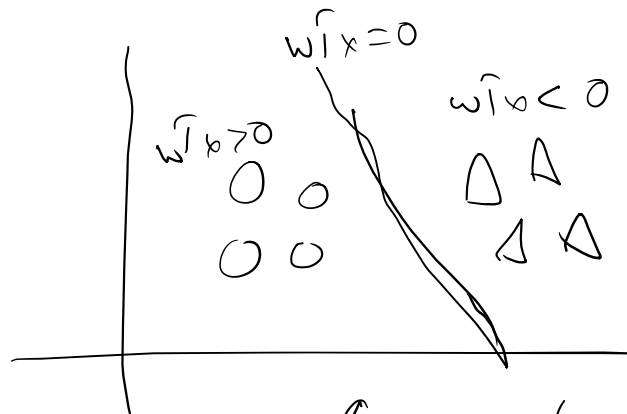


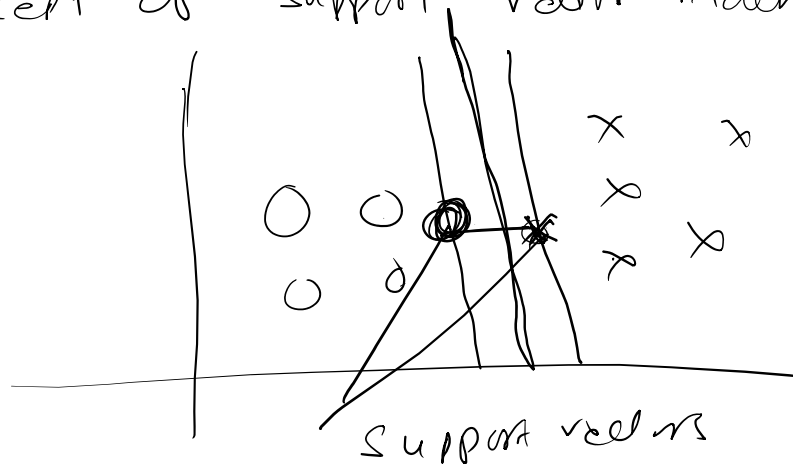
① SVM (Support Vector Machine)

Linear SVM



Simple linear classifier

The Concept of Support Vector Machine



$a_1 + c_1 + c_2 = 0$
 $a_1 + a_2 + c_1 = 1$
 $a_1 + c_1 + c_2 = -1$

Support vector Machine algorithm:

$$w^T x + b = 0 \text{ hyperplane}$$

$$w^T x + b > 0 \rightarrow \text{predict class 1}$$

$$w^T x + b < 0 \rightarrow \text{predict class 0}$$

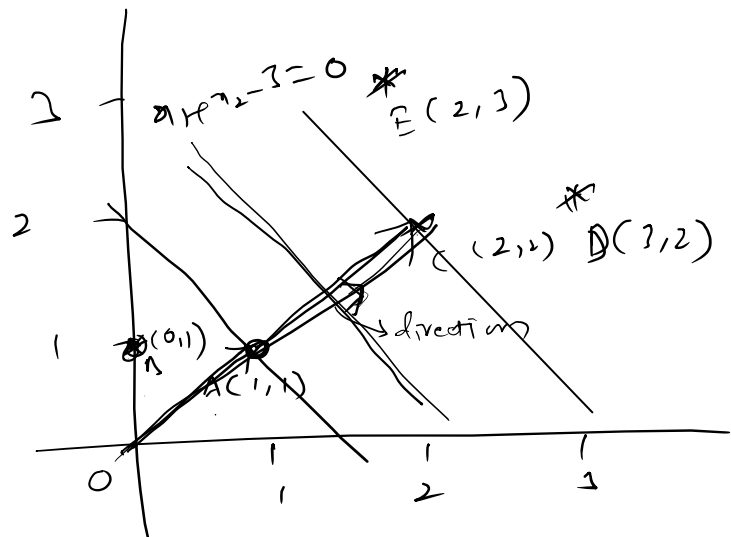
$$w = (w_1, w_2)$$

SVM margin width = $\frac{2}{\|w\|}$ $(w = (0, 1-2))$

subject to $y_i (w^T x + b) \geq 1$

$y_i (w^T x + b) = 1$

x_1	x_2	y
1	1	-1
0	1	-1
2	2	+1
3	2	+1
2	3	+1



A(1,1) B(0,1)
-1

C(2,2) D(3,2) E(2,3)
+1

~~closest~~

closest opposing pair A(1,1) C(2,2)

$dist(A, C) = \sqrt{(2-1)^2 + (2-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.414$

$dist(B, C) = \sqrt{(2-0)^2 + (2-1)^2} = \sqrt{2^2 + 1} = \sqrt{4+1} = \sqrt{5} = 2.23$

$dist(A, D) = \sqrt{(3-1)^2 + (2-1)^2} = \sqrt{2^2 + 1} = \sqrt{4+1} = \sqrt{5} = 2.23$

A(1,1)
-1

C(2,2)
+1

our support vectors

direction: $C - A = \underline{(2,2)} - \underline{(1,1)} = \underline{(1,1)}$

$w = k \cdot (1, 1)$ scalar k will find the value of k

each support vector must satisfy

$$y_i (w \cdot x + b) = 1 \quad \text{with } w = \underline{(k, k)}$$

Support vector $(\underline{2}, \underline{2})$, $y = +1$

$$(+1) (k \cdot 2 + k \cdot 2 + b) = 1$$

$$2k + 2k + b = 1$$

$$4k + b = 1 \quad \text{--- ①}$$

Support vector $A(\underline{1}, \underline{1})$, $y = -1$

$$(-1) (k \cdot 1 + k \cdot 1 + b) = 1$$

$$-(2k + b) = 1$$

$$2k + b = -1 \quad \text{--- ②}$$

from ① & ②

$$4k + b = 1 \quad \text{--- ①}$$

$$2k + b = -1 \quad \text{--- ②}$$

$$2k = 2$$

$$k = \underline{1}$$

$$4k + b = 1 \Rightarrow 4(1) + b = 1$$

$$4 + b = 1$$

$$b = 1 - 4$$

$$b = -3$$

$$w = (1, 1) \Rightarrow w = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad b = -3 \quad \checkmark$$

$$w^T x + b = 0 \rightarrow 1) \pi_1 + (1) \pi_2 - 3 = 0$$

$$\pi_1 + \pi_2 - 3 = 0$$

every point $\pi_1 + \pi_2 > 3$ in class + $\pi_1 + \pi_2 = 3$

$\pi_1 + \pi_2 < 3$ in class -

A (1,1)

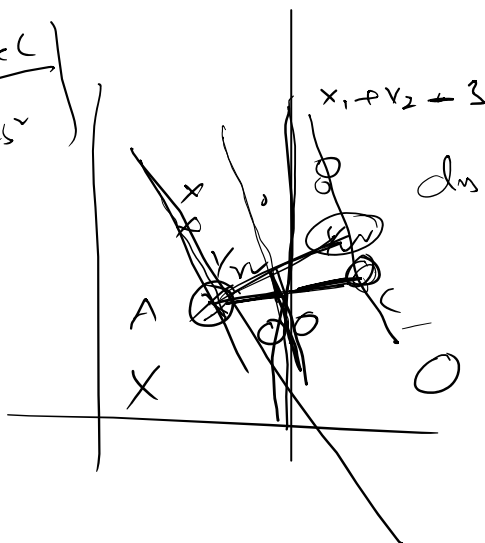
C (2,2)

$$\pi_1 + \pi_2 = 2$$

$$\pi_1 + \pi_2 = 4$$

$$\pi_1 + \pi_2 = 2$$

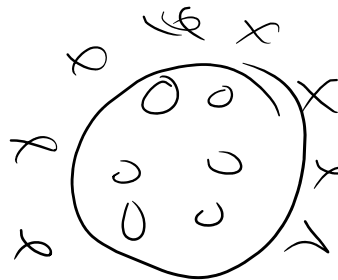
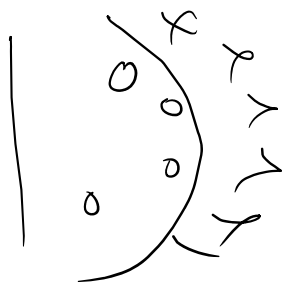
$$\frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$$



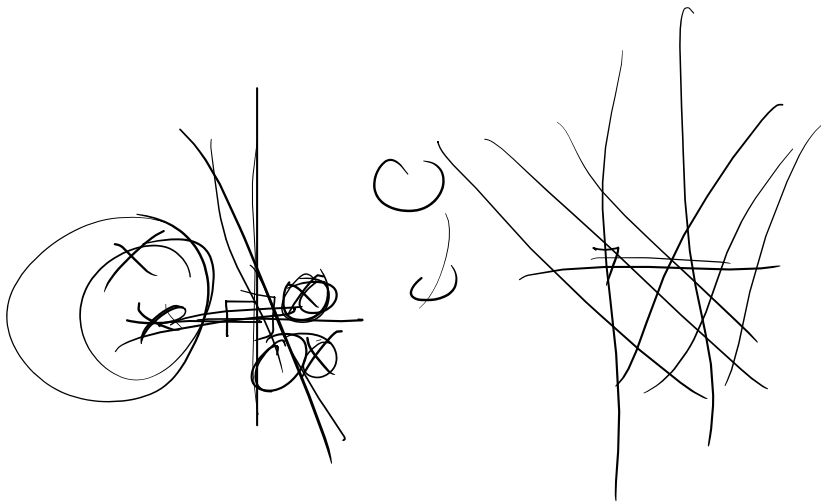
$$\text{distance} = \frac{|x_1 + x_2 - 3|}{\sqrt{1^2 + 1^2}} = \frac{|\pi_1 + \pi_2 - 3|}{\sqrt{2}}$$

$$A(1,1) = \left| \frac{1+1-3}{\sqrt{2}} \right| = \frac{1}{\sqrt{2}}$$

$$C(2,2) = \left| \frac{2+2-3}{\sqrt{2}} \right| = \frac{1}{\sqrt{2}}$$



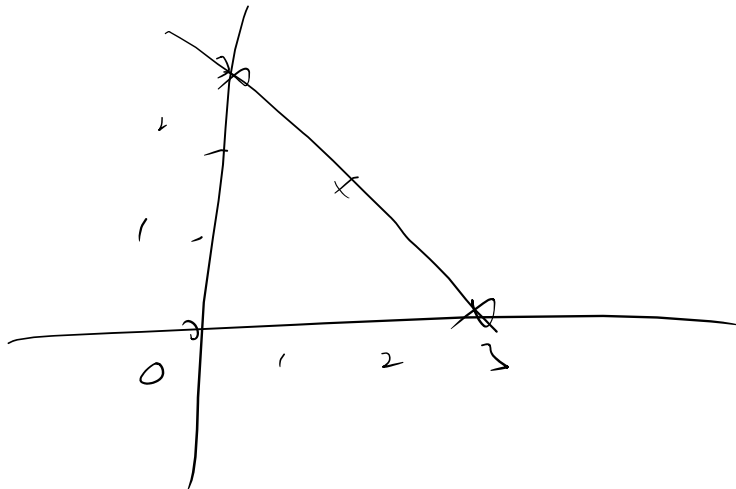
non-linear SVM



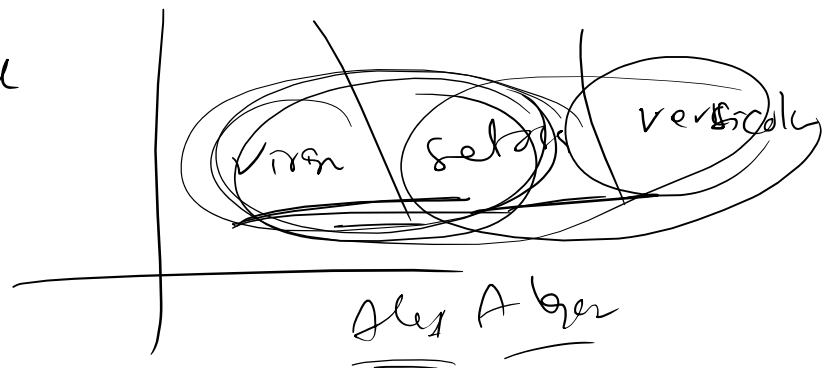
$$n_1 + n_2 = 3$$

$$n_1 = 0 \quad n_2 = 3$$

$$n_1 = 3 \quad n_2 = 0$$



one to one classifier



linear classifier $\rightarrow$ SUM (linear)

